

CCSU BCA Semester 2 – Discrete Mathematics

2026 Exam-Oriented Important Topics + Short Notes

1. Logic & Propositions

Short Note: A proposition is a statement that is either true or false. Learn logical connectives AND, OR, NOT, implication and equivalence; construct truth tables.

Exam focus: Definition + key points + example/diagram/program where applicable.

2. Sets & Relations

Short Note: Learn set operations, Cartesian product and relation properties: reflexive, symmetric, antisymmetric and transitive.

Exam focus: Definition + key points + example/diagram/program where applicable.

3. Functions

Short Note: Study one-one, onto and bijective functions, composition and inverse functions.

Exam focus: Definition + key points + example/diagram/program where applicable.

4. Counting

Short Note: Prepare permutation, combination, pigeonhole principle and basic counting rules.

Exam focus: Definition + key points + example/diagram/program where applicable.

5. Recurrence Relations

Short Note: A recurrence defines a sequence using earlier terms. Learn basic formation and simple solution techniques used in algorithm analysis.

Exam focus: Definition + key points + example/diagram/program where applicable.

6. Graphs

Short Note: Learn vertices, edges, degree, paths, cycles, connected graphs, trees and basic graph representations.

Exam focus: Definition + key points + example/diagram/program where applicable.

7. Trees

Short Note: Study rooted trees, binary trees and basic properties/applications.

Exam focus: Definition + key points + example/diagram/program where applicable.

■ Important Questions

- Explain **Logic & Propositions** with suitable example/diagram/formula/program.
- Explain **Sets & Relations** with suitable example/diagram/formula/program.
- Explain **Functions** with suitable example/diagram/formula/program.
- Explain **Counting** with suitable example/diagram/formula/program.
- Explain **Recurrence Relations** with suitable example/diagram/formula/program.
- Explain **Graphs** with suitable example/diagram/formula/program.
- Explain **Trees** with suitable example/diagram/formula/program.

Note: These are revision notes aligned to the CCSU BCA curriculum structure available for 2025–26. Check your college's paper combination/electives before the exam.